

LE\_BMS\_A\_V2 Fabrication Document

Layer Stack Legend

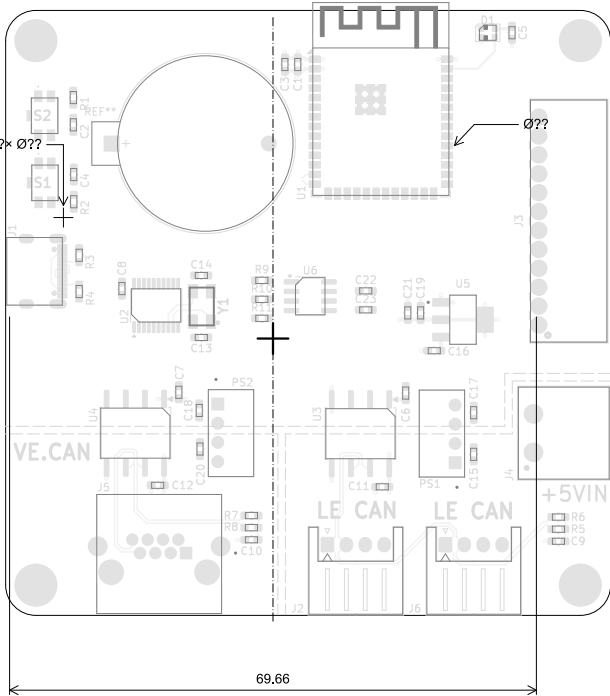
|  | Material     | Layer         | Thickness        | Dielectric    | Type        | Gerber |
|--|--------------|---------------|------------------|---------------|-------------|--------|
|  | F,Paste      |               |                  |               | Paste Mask  |        |
|  | F,Silkscreen |               |                  |               | Legend      | GBR    |
|  | F,Mask       |               | 0,02mm           | Solder Resist | Solder Mask | GBR    |
|  | Copper       | L1 (Sig, PWR) | 0,07mm (2,00oz)  |               | Signal      | GBR    |
|  | Prepreg      |               | 0,1mm            | FR4_7628      | Dielectric  |        |
|  | Copper       | L2 (GND)      | 0,035mm (1,00oz) |               | Plane       | GBR    |
|  | Core         |               | 1,21mm           | FR4           | Dielectric  |        |
|  | Copper       | L3 (Sig, PWR) | 0,035mm (1,00oz) |               | Signal      | GBR    |
|  | Prepreg      |               | 0,1mm            | FR4_7628      | Dielectric  |        |
|  | Copper       | L6 (Sig, PWR) | 0,07mm (2,00oz)  |               | Signal      | GBR    |
|  | B,Mask       |               | 0,02mm           | Solder Resist | Solder Mask | GBR    |
|  | B,Silkscreen |               |                  |               | Legend      | GBR    |
|  | B,Paste      |               |                  |               | Paste Mask  |        |

Total thickness: 1,66mm  
Note: external layer thicknesses are specified after plating

Impedance Table

| Transmission Line              | Impedance [ohms] | Tolerance [ohms] | Layer | Trace Width [mm] | Gap [mm] | Ref. Layers |
|--------------------------------|------------------|------------------|-------|------------------|----------|-------------|
| Edge-Coupled Coated Microstrip | 100              | ±10 %            | L1    | 0,2032           | 0,28     | L2          |

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge\_Cuts.GBR" SUFFIX.  
  
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.  
  
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

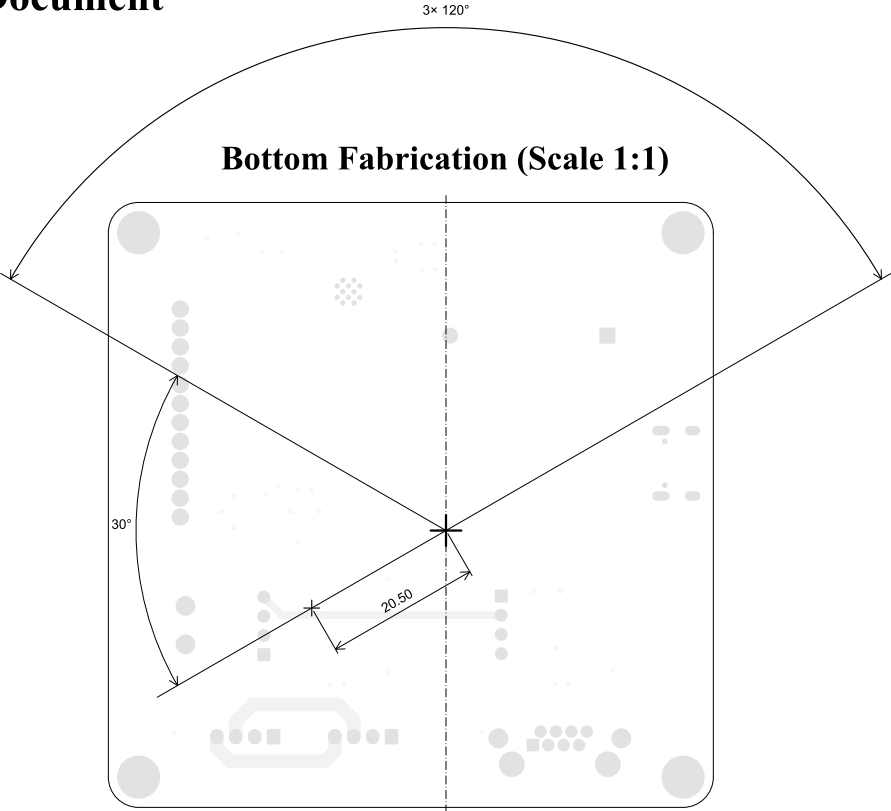
A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.  
B. Tg 170 C OR EQUIVALENT.  
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY COMPANY.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE80.000 × 80.000 mm  
BOARD THICKNESS1.660 mm  
TRACE WIDTH0.200 mm  
TRACE TO TRACE0.200 mm  
MIN. HOLE (PTH)0.200 mm  
MIN. HOLE (NPTH)0.650 mm  
ANNULAR RING0.150 mm  
COPPER TO HOLE0.254 mm  
COPPER TO EDGE0.250 mm  
HOLE TO HOLE0.254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

|                             |                       |              |           |             |                |           |
|-----------------------------|-----------------------|--------------|-----------|-------------|----------------|-----------|
|                             | Comments:             | Company:     |           | Variant:    | Git Hash:      |           |
|                             |                       | Company Name |           | PRELIMINARY | 409ff67        |           |
|                             |                       | Board Name:  |           |             | Project Name:  |           |
|                             |                       | LE_BMS_A_V2  |           |             | LE_V2          |           |
|                             | Sheet Title:          | File Name:   |           | Designer:   | Date:          | Revision: |
| Top Fabrication (Scale 1:1) | LE_BMS_A_V2.kicad_pcb |              | Author    | 2024-04-13  | + (Unreleased) |           |
| Sheet Path:                 |                       |              | Reviewer: | Size:       | Sheet:         |           |
|                             |                       |              |           | A4          | 1 of 10        |           |

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All dimensions are in millimeters unless otherwise specified.

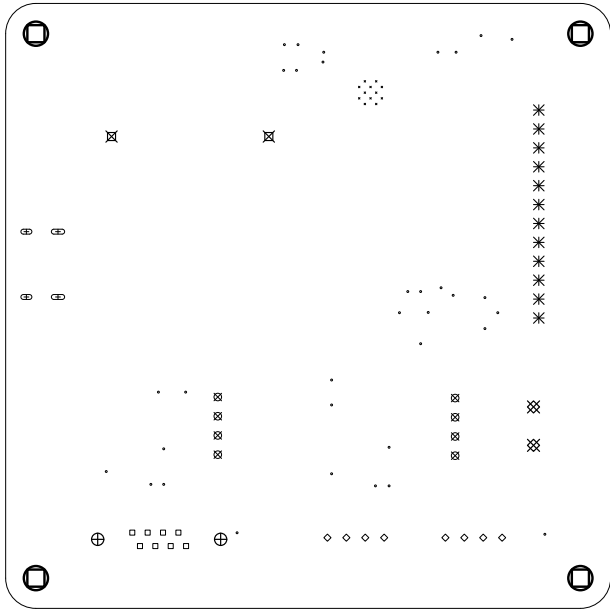
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|  |                                | Company Name          |           | PRELIMINARY   | 409ff67        |
|  |                                | Board Name:           |           | Project Name: |                |
|  |                                | LE_BMS_A_V2           |           | LE_V2         |                |
|  | Sheet Title:                   | File Name:            | Designer: | Date:         | Revision:      |
|  | Bottom Fabrication (Scale 1:1) | LE_BMS_A_V2.kicad_pcb | Author    | 2024-04-13    | + (Unreleased) |
|  | Sheet Path:                    |                       | Reviewer: | Size:         | Sheet:         |
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LE\_BMS\_A\_V2 Fabrication Document

Drill Table

| Symbol   | Count | Hole Size           | Plated | Hole Shape | Drill Layer Pair              | Hole Type |
|----------|-------|---------------------|--------|------------|-------------------------------|-----------|
| ×        | 12    | 0,20mm (7,87mils)   | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| ○        | 34    | 0,25mm (9,84mils)   | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Via       |
| +        | 4     | 0,65mm (25,59mils)  | PTH    | Slot       | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| □        | 8     | 0,94mm (37,01mils)  | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| ◇        | 8     | 0,95mm (37,40mils)  | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| ⊗        | 8     | 1,00mm (39,37mils)  | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| ※        | 12    | 1,40mm (55,12mils)  | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| ⊗        | 2     | 1,50mm (59,06mils)  | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| ⊗        | 2     | 1,60mm (62,99mils)  | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| ⊕        | 2     | 1,62mm (63,78mils)  | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| ○        | 4     | 3,20mm (125,98mils) | PTH    | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Pad       |
| Total 96 |       |                     |        |            |                               |           |

Drill Drawing L1 - L4 (Scale 1:1)



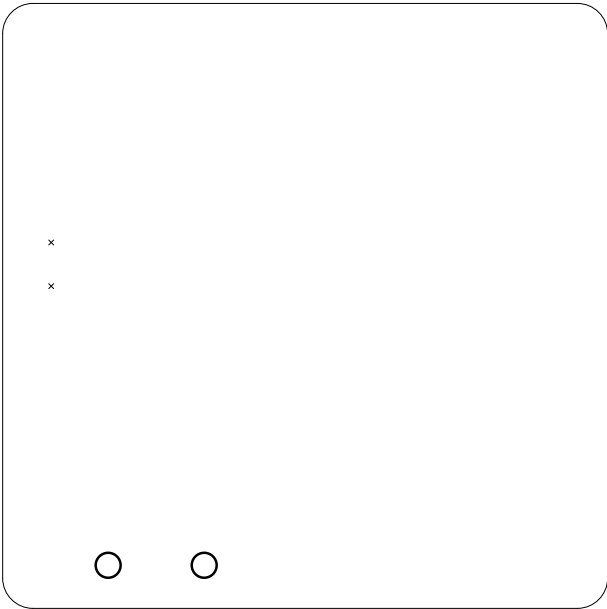
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|  | Comments:                               | Company:<br>Company Name            |                     | Variant:<br>PRELIMINARY | Git Hash:<br>409ff67        |
|  |                                         | Board Name:<br>LE_BMS_A_V2          |                     | Project Name:<br>LE_V2  |                             |
|  | Sheet Title:<br>Drill Drawing (L1 - L4) | File Name:<br>LE_BMS_A_V2.kicad_pcb | Designer:<br>Author | Date:<br>2024-04-13     | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                             |                                     | Reviewer:           | Size:<br>A4             | Sheet:<br>3 of 10           |

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Drill Table

| Symbol | Count   | Hole Size           | Plated | Hole Shape | Drill Layer Pair              | Hole Type  |
|--------|---------|---------------------|--------|------------|-------------------------------|------------|
| X      | 2       | 0,85mm (25,59mils)  | NPTH   | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Mechanical |
| O      | 2       | 3,30mm (129,92mils) | NPTH   | Round      | L1 (Sig, PWR) - L6 (Sig, PWR) | Mechanical |
|        | Total 4 |                     |        |            |                               |            |

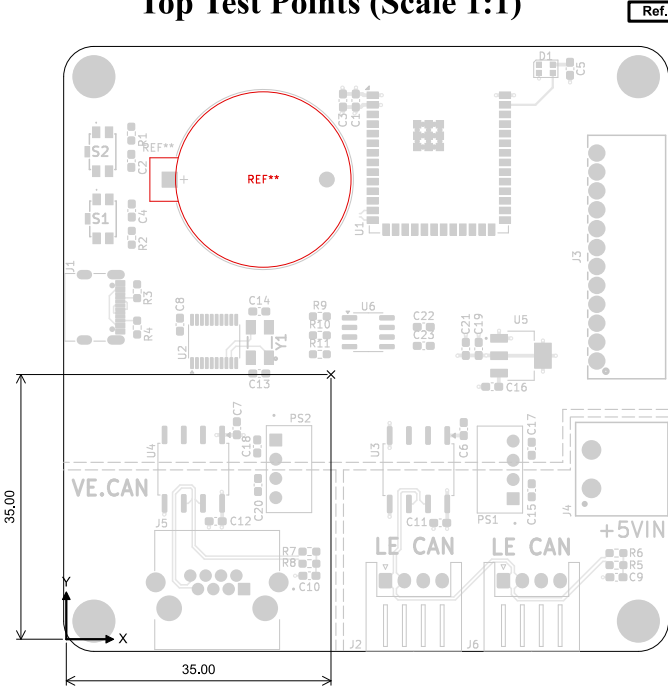
Drill Drawing L1 - L4 (Scale 1:1)



|                         |                       |              |            |                |               |           |  |
|-------------------------|-----------------------|--------------|------------|----------------|---------------|-----------|--|
|                         | Comments:             | Company:     |            | Variant:       |               | Git Hash: |  |
|                         |                       | Company Name |            | PRELIMINARY    |               | 409ff67   |  |
|                         | Sheet Title:          | Board Name:  |            |                | Project Name: |           |  |
|                         |                       | LE_BMS_A_V2  |            |                | LE_V2         |           |  |
| Drill Drawing (L1 - L4) | File Name:            | Designer:    | Date:      | Revision:      |               |           |  |
|                         | LE_BMS_A_V2.kicad_pcb | Author       | 2024-04-13 | + (Unreleased) |               |           |  |
| Sheet Path:             |                       | Reviewer:    | Size:      | Sheet:         |               |           |  |
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LE\_BMS\_A\_V2 Fabrication Document

Top Test Points (Scale 1:1)



| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|

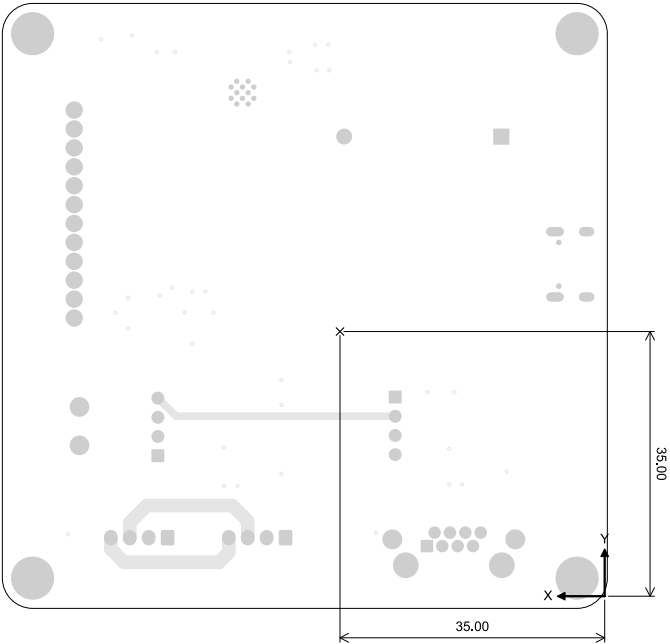
| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|

All dimensions are in millimeters unless otherwise specified.

|  |                             |                       |           |               |                |
|--|-----------------------------|-----------------------|-----------|---------------|----------------|
|  | Comments:                   | Company:              |           | Variant:      | Git Hash:      |
|  |                             | Company Name          |           | PRELIMINARY   | 409ff67        |
|  |                             | Board Name:           |           | Project Name: |                |
|  |                             | LE_BMS_A_V2           |           | LE_V2         |                |
|  | Sheet Title:                | File Name:            | Designer: | Date:         | Revision:      |
|  | Top Test Points (Scale 1:1) | LE_BMS_A_V2.kicad_pcb | Author    | 2024-04-13    | + (Unreleased) |
|  | Sheet Path:                 |                       | Reviewer: | Size:         | Sheet:         |
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Bottom Test Points (Scale 1:1)



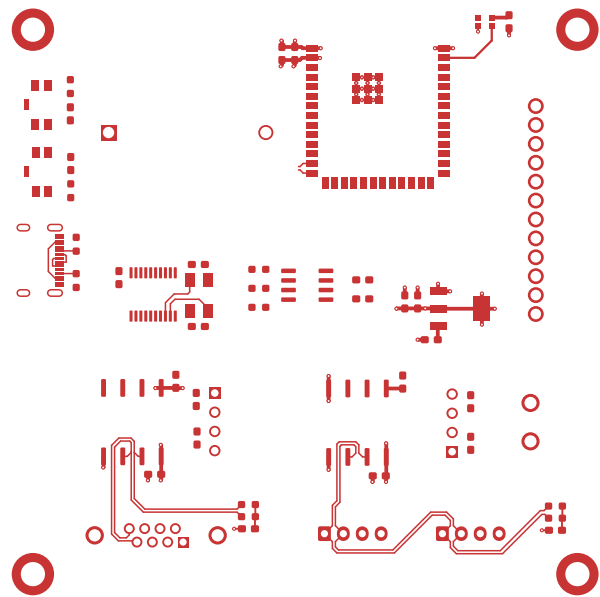
| Ref. | Net | X [mm] | Y [mm] |
|------|-----|--------|--------|
|------|-----|--------|--------|

All dimensions are in millimeters unless otherwise specified.

|  |                                                |                                     |                     |                         |                             |
|--|------------------------------------------------|-------------------------------------|---------------------|-------------------------|-----------------------------|
|  | Comments:                                      | Company:<br>Company Name            |                     | Variant:<br>PRELIMINARY | Git Hash:<br>409ff67        |
|  |                                                | Board Name:<br>LE_BMS_A_V2          |                     | Project Name:<br>LE_V2  |                             |
|  | Sheet Title:<br>Bottom Test Points (Scale 1:1) | File Name:<br>LE_BMS_A_V2.kicad_pcb | Designer:<br>Author | Date:<br>2024-04-13     | Revision:<br>+ (Unreleased) |
|  | Sheet Path:                                    |                                     | Reviewer:           | Size:<br>A4             | Sheet:<br>6 of 10           |

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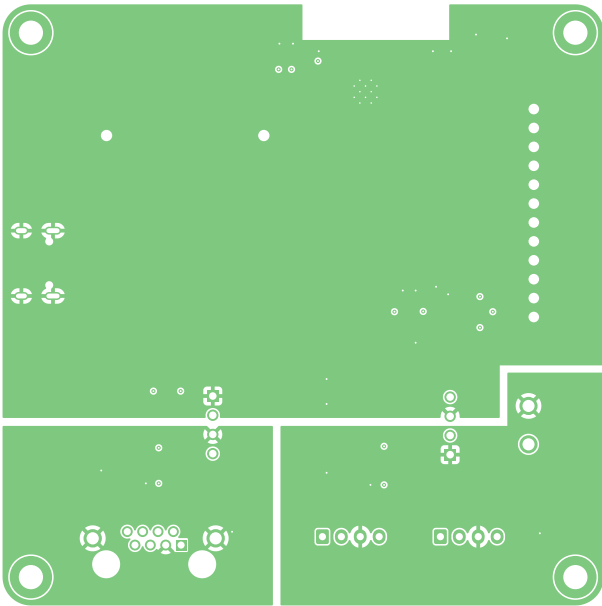
L1 (Sig, PWR) (Scale 1:1)



|  |                           |                       |           |               |                |
|--|---------------------------|-----------------------|-----------|---------------|----------------|
|  | Comments:                 | Company:              |           | Variant:      | Git Hash:      |
|  |                           | Company Name          |           | PRELIMINARY   | 409ff67        |
|  |                           | Board Name:           |           | Project Name: |                |
|  |                           | LE_BMS_A_V2           |           | LE_V2         |                |
|  | Sheet Title:              | File Name:            | Designer: | Date:         | Revision:      |
|  | L1 (Sig, PWR) (Scale 1:1) | LE_BMS_A_V2.kicad_pcb | Author    | 2024-04-13    | + (Unreleased) |
|  | Sheet Path:               |                       | Reviewer: | Size:         | Sheet:         |
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L2 (GND) (Scale 1:1)

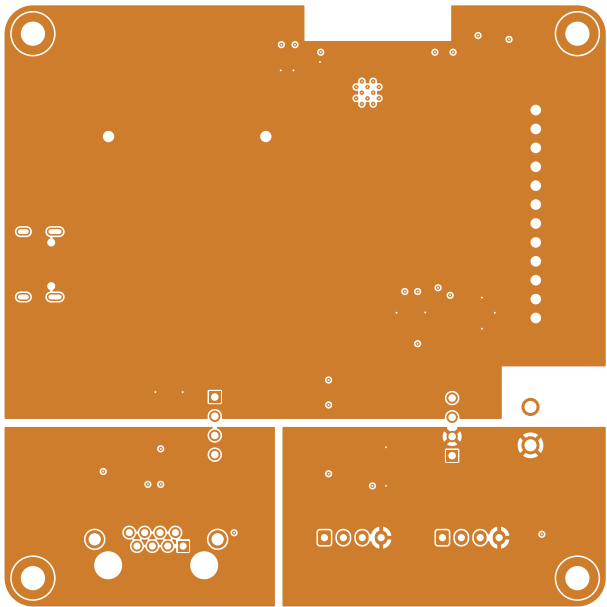


|  |              |                       |           |               |                |
|--|--------------|-----------------------|-----------|---------------|----------------|
|  | Comments:    | Company:              |           | Variant:      | Git Hash:      |
|  |              | Company Name          |           | PRELIMINARY   | 409ff67        |
|  | Sheet Title: | Board Name:           |           | Project Name: |                |
|  |              | L2 (GND) (Scale 1:1)  |           | LE_V2         |                |
|  | Sheet Path:  | File Name:            | Designer: | Date:         | Revision:      |
|  |              | LE_BMS_A_V2.kicad_pcb | Author    | 2024-04-13    | + (Unreleased) |
|  |              |                       | Reviewer: | Size:         | Sheet:         |
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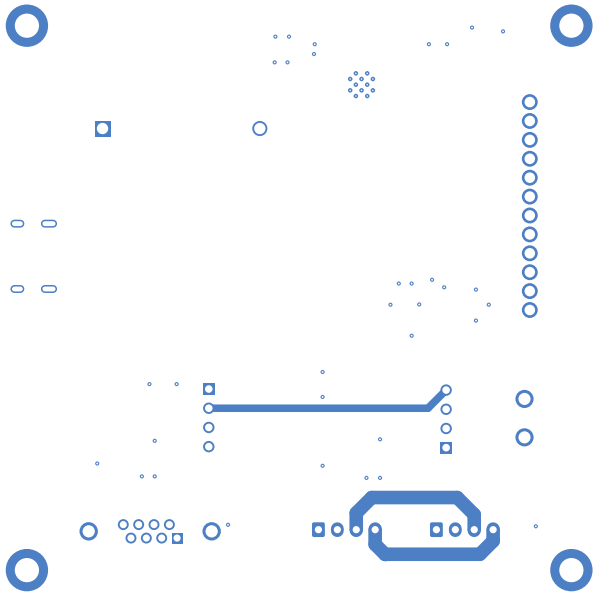
L3 (Sig, PWR) (Scale 1:1)



|  |              |                           |           |               |                |
|--|--------------|---------------------------|-----------|---------------|----------------|
|  | Comments:    | Company:                  |           | Variant:      | Git Hash:      |
|  |              | Company Name              |           | PRELIMINARY   | 409ff67        |
|  | Sheet Title: | Board Name:               |           | Project Name: |                |
|  |              | L3 (Sig, PWR) (Scale 1:1) |           | LE_V2         |                |
|  | Sheet Path:  | File Name:                | Designer: | Date:         | Revision:      |
|  |              | LE_BMS_A_V2.kicad_pcb     | Author    | 2024-04-13    | + (Unreleased) |
|  |              | Reviewer:                 |           | Size:         | Sheet:         |
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L6 (Sig, PWR) (Scale 1:1)



|  |              |                           |           |               |                |
|--|--------------|---------------------------|-----------|---------------|----------------|
|  | Comments:    | Company:                  |           | Variant:      | Git Hash:      |
|  |              | Company Name              |           | PRELIMINARY   | 409ff67        |
|  | Sheet Title: | Board Name:               |           | Project Name: |                |
|  |              | L6 (Sig, PWR) (Scale 1:1) |           | LE_V2         |                |
|  | Sheet Path:  | File Name:                | Designer: | Date:         | Revision:      |
|  |              | LE_BMS_A_V2.kicad_pcb     | Author    | 2024-04-13    | + (Unreleased) |
|  |              |                           | Reviewer: | Size:         | Sheet:         |
|  |              |                           |           | A4            | 10 of 10       |